· **Azi** – continuăm cu aceleași rep. tip discret
- continuu
- fc. repartiție
- fc. caracteristică, fc. generatoare.
- vectori bidimensionali
- LNM (legi ale numerelor mari)] elem. la stat. mat
- TLC (teorema limită centrală)]

· X v.a. discret $X: \begin{pmatrix} x_1 & x_2 & \dots & x_k & \dots \\ p_1 & p_2 & \dots & p_k & \dots \end{pmatrix}$

$p_i \geq 0 \ \forall i$, $\sum_{i \geq 1} p_i = 1$

· val medie $M(X) = \sum x_k p_k$

· dispers $D^2(X) = M(X^2) - (M(X))^2$

$= M(X - M(X))^2$.

$X^2: \begin{pmatrix} x_1^2 & x_2^2 & \dots & x_k^2 & \dots \\ p_1 & p_2 & \dots & p_k & \dots \end{pmatrix}$

$X - m: \begin{pmatrix} x_1 - m & x_2 - m & \dots & x_k - m & \dots \\ p_1 & p_2 & & p_k & \dots \end{pmatrix}$

abaterea medie pătratică $\sqrt{D^2(X)} = \sigma(X)$

· X v.a absolut continuă – pp. că există o densitate $f(x)$:

$f(x) \geq 0 \quad \forall x$

$\int_{-\infty}^{\infty} f(x)\,dx = 1$

Val medie:

$m = M(X) = \int_{-\infty}^{\infty} x f(x)\,dx$

$D^2(X) = \int_{-\infty}^{\infty} (x - m)^2 f(x)\,dx$

$= M(X^2) - (M(X))^2$

$M(X^2) = \int_{-\infty}^{\infty} x^2 f(x)\,dx$

- am făcut uniforme bernoulliane repartiții binomială geometrică.

5) Repartiția binomial negativă (generalizează la geometrică)

$$p_{n/k} = C_{k-1}^{n-1} p^n q^{k-n}, \quad k \geq n, \quad q = 1-p, \quad p \in (0,1)$$

$k+1$ $k+2$ ....

X : $\begin{pmatrix} 0 1 \dots k & \dots \\ p_{1/k} & \dots \end{pmatrix}$

$$M(X) = \frac{n}{p}, \quad D^2(X) = \frac{nq}{p^2}$$

6) Repartiția hipergeometrică (= schema bilei neîntoarse)

a = albe
b = negre
n extrageri / nu punem bila la loc

$$p_n(k) = \frac{C_a^k C_b^{n-k}}{C_{a+b}^n}$$

$\max(0, n-b) \leq k \leq \min(n, a)$

k_1 k_2

k_1 k_1+1 ... k ... k_2

X : $\begin{pmatrix} k_1 & k_1+1 & \dots & k & \dots & k_2 \\ & & p_n(k) & & & \end{pmatrix}$

$$M(X) = \frac{na}{a+b}, \quad D^2(X) = \frac{nab(a+b-n)}{(a+b)^2(a+b-1)}$$

7) Repartiția Poisson

$\lambda > 0$ $\quad p_n(k,\lambda) = e^{-\lambda} \cdot \frac{\lambda^k}{k!}$, $k \geq 0$.

$\frac{1}{e} + \frac{1}{e} + \frac{1}{2}e =$

$$X : \begin{pmatrix} 0 & 1 & 2 & \cdots & k & \cdots \\ e^{-\lambda} & \lambda e^{-\lambda} & \frac{\lambda^2}{2} e^{-\lambda} & \cdots & e^{-\lambda}\frac{\lambda^k}{k!} & \cdots \end{pmatrix}$$

0	1	e	...
e^{-1}	e^{-1}	$\frac{e^{-1}}{2}$	...

Știm că $e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \cdots + \frac{x^k}{k!} + \cdots = \sum_{k \geq 0} \frac{x^k}{k!}$

$$\sum_{k=0}^{\infty} p_n(k,\lambda) = \sum_{k=0}^{\infty} e^{-\lambda} \cdot \frac{\lambda^k}{k!} = e^{-\lambda} \cdot \underbrace{\sum_{k=0}^{\infty} \frac{\lambda^k}{k!}}_{= e^{\lambda}} =$$

$$= e^{-\lambda} \cdot e^{\lambda} = e^{-\lambda+\lambda} = e^0 = 1.$$

evident $p_n(k,\lambda) \geq 0$

$$M(X) = \sum_{k=0}^{\infty} k \cdot e^{-\lambda} \cdot \frac{\lambda^k}{k!} = e^{-\lambda} \cdot \sum_{k=1}^{\infty} k \cdot \frac{\lambda \lambda^{k-1}}{k \cdot (k-1)!} =$$

$$= e^{-\lambda} \cdot \lambda \cdot \underbrace{\sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!}}_{= e^{\lambda}} = \underbrace{e^{-\lambda} \cdot e^{\lambda}}_{=1} \cdot \lambda = \lambda$$

$$D^2(X) = M(X^2) - \underbrace{(M(X))^2}_{= \lambda} = M(X^2) - \lambda^2$$

$$M(X^2) = \sum_{k=0}^{\infty} k^2 \cdot e^{-\lambda} \cdot \frac{\lambda^k}{k!} = e^{-\lambda} \sum_{k=0}^{\infty} \left[k(k-1) + k \right] \cdot \frac{\lambda^k}{k!}$$

$$= e^{-\lambda} \left[\sum_{k=2}^{\infty} k(k-1) \cdot \frac{\lambda^2 \cdot \lambda^{k-2}}{k(k-1)(k-2)!} + \sum_{k=1}^{\infty} k \cdot \frac{\lambda \lambda^{k-1}}{k \cdot (k-1)!} \right]$$

$$= e^{-\lambda} \cdot \lambda^2 \cdot \underbrace{\sum_{k=2}^{\infty} \frac{\lambda^{k-2}}{(k-2)!}}_{= e^{\lambda}} + e^{-\lambda} \cdot \lambda \cdot \underbrace{\sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!}}_{= e^{\lambda}} =$$

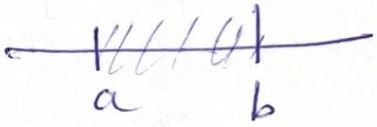
$$= \underbrace{e^{-\lambda} \cdot e^{\lambda}}_{1} \cdot \lambda^2 + \underbrace{e^{-\lambda} \cdot e^{\lambda}}_{1} \cdot \lambda = \lambda^2 + \lambda$$

$$D^2(X) = M(X^2) - (M(X))^2 = \lambda^2 + \lambda - \lambda^2 = \lambda$$

Repartitii continue

Rep. uniformă pe $[a,b]$.

$$f(x)=\begin{cases}\frac{1}{b-a}, & x\in[a,b]\\ 0, & x\notin[a,b]\end{cases}$$



$f\geq 0$ ·
$$\int_{-\infty}^{\infty} f(x)dx=\int_{-\infty}^{a}\underbrace{f(x)dx}_{0}+\int_{a}^{b}\underbrace{f(x)dx}_{\frac{1}{b-a}}+\int_{b}^{\infty}\underbrace{f(x)dx}_{0}=$$

$$=\int_a^b \frac{1}{b-a}dx=\frac{1}{b-a}\int_a^b dx=\frac{1}{b-a}\,x\Big|_a^b=$$

$$=\frac{1}{b-a}(b-a)=1$$

$$M(X)=\int_{-\infty}^{\infty} x f(x)\,dx=\int_a^b x\cdot\frac{1}{b-a}dx=\frac{1}{b-a}\int_a^b x\,dx=$$

$$=\frac{1}{b-a}\,\frac{x^2}{2}\Big|_a^b=\frac{1}{2(b-a)}(b^2-a^2)=\frac{(b-a)(b+a)}{2(b-a)}=\frac{a+b}{2}$$

$$D^2(X)=M(X^2)-(M(X))^2$$

$$M(X^2)=\int_a^b x^2\underbrace{f(x)}_{\frac{1}{b-a}}dx=\frac{x^3}{3}\Big|_a^b\cdot\frac{1}{b-a}=\frac{1}{b-a}\cdot\frac{b^3-a^3}{3}=$$

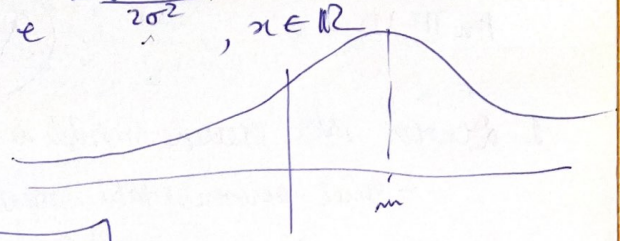
$$=\frac{1}{b-a}\cdot(b-a)\,\frac{(a^2+ab+b^2)}{3}=\frac{a^2+ab+b^2}{3}$$

$$D^2(X)=\frac{a^2+ab+b^2}{3}-\frac{(a+b)^2}{4}=\frac{4a^2+4ab+4b^2-3a^2-3b^2-6ab}{12}$$

$$=\frac{a^2-2ab+b^2}{12}=\frac{(a-b)^2}{12}$$

2) Repartiția normală

$X: \quad f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-m)^2}{2\sigma^2}}, \quad x \in \mathbb{R}$



$f(x; m, \sigma^2)$

m, σ^2

$$M(X) = m$$
$$D^2(X) = \sigma^2$$

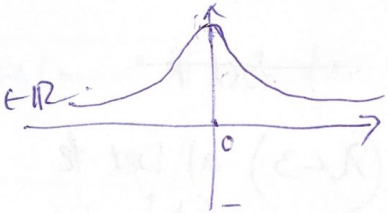
Falsum $\int_{-\infty}^{\infty} e^{-\frac{t^2}{2}} dt = \sqrt{2\pi}$

forma redusă 0,1.

$m = 0, \quad \sigma^2 = 1.$

$f(x) = \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{x^2}{2}}, \quad x \in \mathbb{R}$

Gauss.



$Y = \frac{X-m}{\sigma}$ $\quad M(Y) = 0, \quad D^2(Y) = 1$, v.a. redusă

3) Repartiția log-normală

v.a. X urmează o repart./lege lognormală dacă

$$f(x) = \frac{1}{\sigma x\sqrt{2\pi}} e^{-\frac{(\ln x - m)^2}{2\sigma^2}}, \quad x > 0.$$

Y v.a. normală $\quad X = e^Y$

$$\begin{cases} M(X) = e^{m+\frac{\sigma^2}{2}} \\ D^2(X) = e^{2m+\sigma^2}(e^{\sigma^2}-1) \end{cases}$$

4) Repartiția beta

X v.a $f(x;m,n) = \begin{cases} \frac{x^{m-1}(1-x)^{n-1}}{B(m,n)} & x \in [0,1] \\ 0 & \text{rest } x \in (-\infty,0) \cup (1,\infty) \end{cases}$

$m, n \in \mathbb{R}_+^*$, $m,n>0$

$B(m,n) = \frac{\Gamma(m)\cdot\Gamma(n)}{\Gamma(m+n)}$

$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} \cdot e^{-x}\, dx$, $\alpha > 0$

$\Gamma(\alpha+1) = \alpha\,\Gamma(\alpha)$

$M(X^r) = \int_0^1 x^r \cdot \frac{x^{m-1}(1-x)^{n-1}}{B(m,n)}\, dx.$

Știm $\int_0^1 x^{m-1}(1-x)^{n-1}\, dx = B(m,n)$

$M(X^r) = \frac{1}{B(m,n)} \int_0^1 x^{(m+r)-1}(1-x)^{n-1} dx = \frac{1}{B(m,n)} B(m+r,n)$

$= \frac{\Gamma(m+r)\cdot\Gamma(n)}{\Gamma(m+r+n)} \cdot \frac{\Gamma(m+n)}{\Gamma(m)\,\Gamma(n)} = \frac{\Gamma(m+r)\cdot\Gamma(m+n)}{\Gamma(m)\cdot\Gamma(m+r+n)}$

$r=1 \Rightarrow M(X) = \frac{\Gamma(m+1)\,\Gamma(m+n)}{\Gamma(m)\,\Gamma(m+n+1)} = \frac{m\,\Gamma(m)\cdot\Gamma(m+n)}{\Gamma(m)\,(m+n)\,\Gamma(m+n)} = \frac{m}{m+n}$

$r=2$ $M(X^2) = \frac{\Gamma(m+2)\cdot\Gamma(m+n)}{\Gamma(m)\,\Gamma(m+n+2)}$

$\Gamma(\alpha+1) = \alpha\,\Gamma(\alpha)$

$\Gamma(m+2) = (m+1)\cdot\Gamma(m+1) = (m+1)\cdot m\,\Gamma(m)$

$\Gamma(m+n+2) = (m+n+1)\,\Gamma(m+n+1) = (m+n+1)\cdot(m+n)\cdot\Gamma(m+n)$

Știm $D^2(X) = M(X^2) - (M(X))^2 = \dots$

5) Repartiția exponențială (negativă)

X v.a. exponențială $f(x;\lambda) = \lambda e^{-\lambda x}$, $x>0$.

($\lambda>0$ parametru)

(!) în literatură: $f(x;\lambda) = \frac{1}{\lambda} e^{-\frac{x}{\lambda}}$, $x>0$

$f \geq 0$ evident

$$\int_0^\infty f\,dx = \int_0^\infty \lambda e^{-\lambda x}\,dx =$$

$$= \lambda \int_0^\infty e^{-\lambda x}\,dx = \lambda \left.\frac{e^{-\lambda x}}{-\lambda}\right|_0^\infty$$

$$= -\left.e^{-\lambda x}\right|_0^\infty = -(0-1) = 1$$

$$f = \begin{cases} \lambda e^{-\lambda x}, & x>0 \\ 0, & x\leq 0. \end{cases}$$

$\int e^x dx = e^x + C$

$\int e^{3x}\,dx = \frac{e^{3x}}{3}$

$\int e^{-3x}\,dx = \frac{e^{-3x}}{-3}$ $\quad (e^u)' = u' e^u$

$$M(X) = \int_{-\infty}^{\infty} x f(x)\,dx = \int_0^\infty x f(x)\,dx = \int_0^\infty x \cdot \lambda e^{-\lambda x}\,dx =$$

$$= \lambda \int_0^\infty \underbrace{x}_{u} \underbrace{e^{-\lambda x}}_{dv} dx = \lambda \left[uv\Big|_0^\infty - \int_0^\infty v\,du \right] =$$

$u' = 1$, $du = dx$; $v = \frac{e^{-\lambda x}}{-\lambda}$

$$= \lambda \left[x \frac{e^{-\lambda x}}{-\lambda}\Big|_0^\infty - \int_0^\infty \frac{e^{-\lambda x}}{-\lambda}\,dx \right] =$$

$$= \lambda \left[-\frac{1}{\lambda}(0-0) + \frac{1}{\lambda} \left.\frac{e^{-\lambda x}}{-\lambda}\right|_0^\infty \right] =$$

$$\lim_{x\to\infty} x e^{-\lambda x} = \lim_{x\to\infty} \frac{x}{e^{\lambda x}} \overset{\infty}{=} \lim_{x\to\infty} \frac{(x)'}{(e^{\lambda x})'} = \lim_{x\to\infty} \frac{1}{\lambda e^{\lambda x}} = \frac{1}{\infty} = 0$$

$$= \lambda \cdot \frac{1}{\lambda^2} \left(e^{-\lambda x}\Big|_0^\infty\right) = -\frac{1}{\lambda}(0-1) = \frac{1}{\lambda}.$$

$$D^2(X) = M(X^2) - (M(X))^2$$

$$M(X^2) = \int_0^{\infty} x^2 f(x)\,dx = \lambda \int_0^{\infty} \underbrace{x^2}_{u} \underbrace{e^{-\lambda x}}_{dv} dx = \frac{2}{\lambda^2}$$

$du = 2x\,dx$

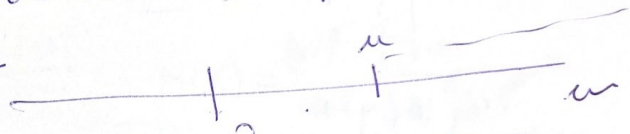
$$D^2(X) = \frac{1}{\lambda^2}$$

6) Repartitia Weibull

$$f(x) = \frac{m}{x_0}\left(\frac{x-u}{x_0}\right)^{m-1} \cdot e^{-\left(\frac{x-u}{x_0}\right)^m}$$

$x_0 > 0$, $m > 0$, $0 \le u \le x < \infty$; pt $x < u$ avem $f(x) = 0$

De obicei $u = 0$



Dacă $x_0 = 1$

$$f(x) = m x^{m-1} e^{-x^m}$$

$$f(x) = \beta x^{\beta-1} e^{-x^\beta}$$

$\frac{1}{x_0} = \lambda$

$m = \beta$

$$f(x) = \lambda\beta (x\lambda)^{\beta-1} e^{-(x\lambda)^\beta}$$

dacă $\beta = 1 \Rightarrow$

$x > 0$

$$f(x) = \lambda e^{-\lambda x}$$ adică

$\exp(\lambda)$ $\exp(\lambda)$.

7) Repartitie Erlang

$$f(x) = \frac{(\lambda k)^k}{\Gamma(k)} x^{k-1} e^{-\lambda k x}$$

pt $k = 1 \Rightarrow f(x) = \lambda e^{-\lambda x}$,

$\exp(\lambda)$

$x > 0$, $\lambda > 0$, $k \ge 1$

$$M(X^r)=\int_0^\infty x^r\cdot f(x)\,dx=\int_0^\infty x^r\cdot\frac{(\lambda k)^k}{\Gamma(k)}\cdot x^{k-1}e^{-\lambda k x}\,dx$$

$$=\frac{(\lambda k)^k}{\Gamma(k)}\int_0^\infty x^{r+k-1}e^{-\lambda k x}\,dx=$$

$\lambda k x=y$

$dy=\lambda k\,dx$

$x=\frac{y}{\lambda k}$

$x=0\Rightarrow y=0$

$x\to\infty\Rightarrow y\to\infty$

$$=\frac{(\lambda k)^k}{\Gamma(k)}\int_0^\infty\left(\frac{y}{\lambda k}\right)^{r+k-1}e^{-y}\cdot\frac{dy}{\lambda k}=$$

$$=\frac{(\lambda k)^k}{\Gamma(k)}\cdot\frac{1}{(\lambda k)^{r+k-1}}\cdot\frac{1}{\lambda k}\left[\int_0^\infty y^{r+k-1}\cdot e^{-y}dy\right]=$$

$$=\frac{(\lambda k)^k}{\Gamma(k)}\cdot\frac{\Gamma(r+k)}{(\lambda k)^{r+k}}\quad;\quad M(X)=\frac{(\lambda k)^k}{\Gamma(k)}\cdot\frac{\Gamma(k+1)}{(\lambda k)^{k+1}}=$$

$r=1$

$$=\frac{k\cdot\Gamma(k)}{\Gamma(k)\cdot(\lambda k)}=\frac{k}{\lambda k}=\frac{1}{\lambda}$$

$r=2$

$$M(X^2)=\frac{(\lambda k)^k}{\Gamma(k)}\cdot\frac{\Gamma(k+2)}{(\lambda k)^{k+2}}=\frac{(k+1)\Gamma(k+1)}{\Gamma(k)(\lambda k)^2}=\frac{k(k+1)\Gamma(k)}{\Gamma(k)\lambda^2k^2}=$$

$$=\frac{k+1}{k\lambda^2}$$

$$\Rightarrow D^2(X)=M(X^2)-(M(X))^2=$$

$$=\frac{k+1}{k\lambda^2}-\left(\frac{1}{\lambda}\right)^2=\frac{k+1-k}{k\lambda^2}=\frac{1}{k\lambda^2}$$

8) Repartiția χ^2

- X v.a rep. $\chi^2(n;\sigma)$; χ^2 cu n grade libertate, param $\sigma>0$

dacă $f(x)=\frac{1}{2^{\frac{n}{2}}\sigma^n\Gamma\left(\frac{n}{2}\right)}x^{\frac{n}{2}-1}\cdot e^{-\frac{x}{2\sigma^2}}$, $x>0$, n natural

- X v.a χ^2 cu n grade libert: $\sigma=1$

$M(X)=n\sigma^2 \qquad M(X^2)=n(n+2)\sigma^4$

$D^2(X)=2n\sigma^4$

(T) $X_1,\dots,X_n$ v.a. independente repartizate $N(0,\sigma^2)$

atunci v.a $Y=X_1^2+X_2^2+\dots+X_n^2 \sim \chi^2(n,\sigma^2)$

9) **Repartiția Student**

X v.a Student : $f(x)=\dfrac{\Gamma\left(\frac{n+1}{2}\right)}{\sqrt{n\pi}\cdot\Gamma\left(\frac{n}{2}\right)}\left(1+\dfrac{x^2}{n}\right)^{-\frac{n+1}{2}}$

$x\in\mathbb{R}$, n natural.

10) **Repartiția Snedecor**

$f(x)=\left(\dfrac{m}{n}\right)^{\frac{m}{2}}\dfrac{\Gamma\left(\frac{m+n}{2}\right)}{\Gamma\left(\frac{m}{2}\right)\cdot\Gamma\left(\frac{n}{2}\right)}x^{\frac{m}{2}-1}\cdot\left(1+\dfrac{m}{n}x\right)^{-\frac{m+n}{2}}$

$x\geq 0$, m,n naturale.

11) **Repartiția Fisher** (Fischer)

$f(x)=2\left(\dfrac{m}{n}\right)^{\frac{m}{2}}\dfrac{\Gamma\left(\frac{m+n}{2}\right)}{\Gamma\left(\frac{m}{2}\right)\Gamma\left(\frac{n}{2}\right)}e^{mx}\cdot\left(1+\dfrac{m}{n}e^{2x}\right)^{-\frac{m+n}{2}}$

12) **Repartiția Cauchy**

$f(x)=\dfrac{1}{\pi}\cdot\dfrac{a}{a^2+(x-b)^2}$, $a>0$, $x\in\mathbb{R}$.

Funcție de repartiție

P, X v.a.

$F_X(x) = P(X < x)$ sau $F_X(x) = P(X \leq x)$.

!

→ dacă v.a. X nu continuă (nu) sunt probleme ⇒ cele două forme coincid pt că $P(X = x_0) = 0$! în caz continuu

În caz discret (nu) coincid!

Caz continuu: Pp. X v.a. absolut cont.

F și atunci avem $f = F'$

Ex $f(x) = \lambda e^{-\lambda x}$, $x > 0$ $\quad X \in exp(\lambda)$

Calculăm $F(x)$

$$F(x) = \int_{-\infty}^{x} f(t)\,dt$$

$$f(x) = \begin{cases} 0, & x \leq 0 \\ \lambda e^{-\lambda x}, & x > 0 \end{cases}$$

$x \leq 0 \Rightarrow F(x) = 0$.

$x > 0 \Rightarrow F(x) = \int_{-\infty}^{x} f(t)\,dt = \int_{-\infty}^{0} f(t)\,dt + \int_{0}^{x} f(t)\,dt$

$= \int_0^x \lambda e^{-\lambda t}\,dt = \lambda \int_0^x e^{-\lambda t}\,dt = \lambda \cdot \frac{e^{-\lambda t}}{-\lambda}\Big|_0^x = -(e^{-\lambda x} - 1)$

$= 1 - e^{-\lambda x}$

$$F(x)=\begin{cases}0, & x\le 0\\ 1-e^{-\lambda x}, & x>0\end{cases}$$

Ex $X\sim exp(\lambda)$

Fie $Y=3X-5$. Să det F_Y și f_Y:

Sol: $F_Y(y)=P(Y\le y)=P(3X-5\le y)=$

$=P\left(X\le \frac{y+5}{3}\right)=F_X\left(\frac{y+5}{3}\right)=\begin{cases}0, & \frac{y+5}{3}\le 0\\ 1-e^{-\lambda\left(\frac{y+5}{3}\right)}, & \frac{y+5}{3}>0\end{cases}$

$=\begin{cases}0, & y\le -5\\ 1-e^{-\lambda\left(\frac{y+5}{3}\right)}, & y>-5\end{cases}$

Ex $X\sim exp(\lambda)$

Să det pt $Y=4-9X^2$ fc. F_Y, f_Y

Sol: $F_Y(y)=P(Y\le y)=P(4-9X^2\le y)=$

$=P(9X^2\ge 4-y)=$

dacă $4-y\le 0\Rightarrow$ **$F_Y(y)=1$. pt $y\ge 4$**

dacă $4-y>0$

$9x^2\ge 4-y$

$x^2\ge \frac{4-y}{9}$.

$x\in\left(-\infty,-\frac{\sqrt{4-y}}{3}\right]\cup\left[\frac{\sqrt{4-y}}{3},+\infty\right)$

$x^2\le 16$

$x\in[-4,4]$.

$x^2\ge 16$.

$x\in(-\infty,-4]\cup[4,\infty)$

$=P\left(X\in\left(-\infty,-\frac{\sqrt{4-y}}{3}\right]\cup\left[\frac{\sqrt{4-y}}{3},+\infty\right)\right)=$

$$= P\Big(\underbrace{X}_{\leq 0} \leq -\frac{\sqrt{4-y}}{3}\Big) + P\Big(\underbrace{X \geq \frac{\sqrt{4-y}}{3}}_{\bar{A}}\Big) =$$

ϕ

$$= \int_{\frac{\sqrt{4-y}}{3}}^{\infty} f(t)\,dt = 1 - P\Big(X \leq \frac{\sqrt{4-y}}{3}\Big)$$

$A = \{X \leq \frac{\sqrt{\ }}{3}\}$

$\bar{A} = \{X \geq \ \}$

$$= 1 - F_X\Big(\frac{\sqrt{4-y}}{3}\Big) = 1 - \begin{cases} 0, & \frac{\sqrt{4-y}}{3} \leq 0 \\ 1 - e^{-\lambda \frac{\sqrt{4-y}}{3}}, & \frac{\sqrt{4-y}}{3} \geq 0 \end{cases}$$

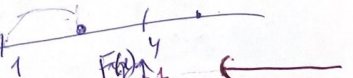
$$= \begin{cases} 1, & y \geq 4 \\ \to e^{\frac{\sqrt{4-y}}{3}}, & y \leq 4. \end{cases}$$

Caz discret

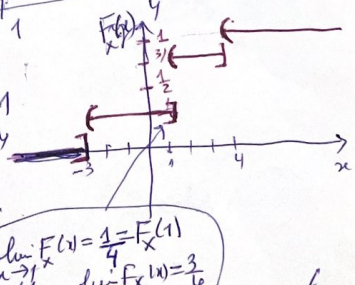
Cazul $F_X(x) = P(X < x)$ continuu la stg

Ex $X : \begin{pmatrix} -3 & 1 & 4 \\ \frac{1}{4} & \frac{2}{4} & \frac{1}{4} \end{pmatrix}$

Să calc. $F_X(x)$ și să facem graficul corespunzător



$$F_X(x) = \begin{cases} 0 & x \leq -3 \\ 0 + \frac{1}{4} & x \in (-3, 1] \\ 0 + \frac{1}{4} + \frac{2}{4} & x \in (1, 4] \\ 0 + \frac{1}{4} + \frac{2}{4} + \frac{1}{4} & x \in (4, \infty) \end{cases} = \begin{cases} 0 & x \leq -3 \\ \frac{1}{4}, & -3 < x \leq 1 \\ \frac{3}{4} & 1 < x \leq 4 \\ 1 & x > 4. \end{cases}$$

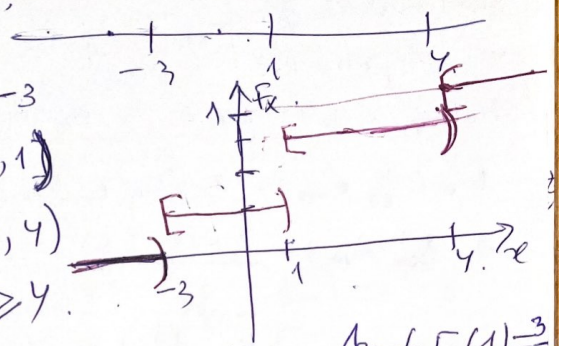


$\lim_{\substack{x \to 1 \\ x < 1}} F_X(x) = \frac{1}{4} = F_X(1)$

$\lim_{\substack{x \to 1 \\ x > 1}} F_X(x) = \frac{3}{4}$.

Cazul $F_X(x) = P(X \leq x)$ cont la dreapta (14)

Ex de mai sus!



$$F_X(x) = \begin{cases} 0 & x < -3 \\ 0 + \frac{1}{4} & x \in [-3, 1) \\ 0 + \frac{1}{4} + \frac{2}{4} & x \in [1, 4) \\ 1 & x \geq 4 \end{cases}$$

$\lim_{\substack{x \to 1 \\ x < 1}} F_X(x) = \frac{1}{4} \neq F_X(1) = \frac{3}{4}$

$= \lim_{\substack{x \to 1 \\ x > 1}} F_X(x)$.

Ex $X : \begpmatrix 4 & -3 & 1 \\ \frac{2}{5} & \frac{2}{5} & \frac{1}{5} \endpmatrix$, $X : \begpmatrix -3 & 1 & 4 \\ \frac{2}{5} & \frac{1}{5} & \frac{2}{5} \endpmatrix$

$F_X(x) = P(X \leq x)$

Funcția generatoare

$G_X(t) = M(e^{tX})$

$G_X : \mathbb{R} \to \mathbb{R}$

caz discret $X : \begpmatrix x_1 & x_2 & \dots & x_k & \dots \\ p_1 & p_2 & \dots & p_k & \dots \endpmatrix$

$G_X(t) = \sum_{k \geq 1}^{\infty} e^{tx_k} \cdot p_k$

caz continuu (abs cont)

$G_X(t) = \int_{-\infty}^{\infty} e^{tx} \cdot f(x)\,dx$

Funcția caracteristică

$\varphi_X(t) = M(e^{itX})$ $\quad i = \sqrt{-1}$

$\varphi_X : \mathbb{R} \to \mathbb{C}$

$\varphi_X(t) = \sum_{k \geq 1} e^{itx_k} \cdot p_k$

$\varphi_X(t) = \int_{-\infty}^{\infty} e^{itx} f(x)\,dx$.

Ex μt discut

$$X:\begin{pmatrix} -3 & 1 & 4 \\ \frac{1}{4} & \frac{2}{4} & \frac{1}{4} \end{pmatrix}$$

$$G_X(t)=e^{-3t}\cdot\frac{1}{4}+e^{1\cdot t}\cdot\frac{2}{4}+e^{4\cdot t}\cdot\frac{1}{4}.$$

~~Def~~ Cf cout exp

$$G_X(t)=\int_{-\infty}^{\infty} e^{tx} f(x)\,dx=\int_0^{\infty} e^{tx}\cdot\lambda e^{-\lambda x}\,dx$$

$$=\lambda\int_0^{\infty} e^{x(t-\lambda)}\,dx=\lambda\cdot\frac{e^{x(t-\lambda)}}{t-\lambda}\Big|_0^{\infty}=\lambda\left(0-\frac{1}{t-\lambda}\right)=\frac{\lambda}{\lambda-t}$$

$$\varphi_X(t)=e^{-3it}\cdot\frac{1}{4}+e^{it}\cdot\frac{2}{4}+e^{4it}\cdot\frac{1}{4}.$$

$$\varphi_X(t)=\int_0^{\infty} e^{itx}\cdot\lambda e^{-\lambda x}\Rightarrow\int_0^{\infty} e^{x(it-\lambda)}\,dx$$

$=$

$(t\neq\lambda)$

$t<\lambda$

Reveniu

$$G_X(t)=\sum_{k\geq 1} e^{t x_k}\cdot p_k$$

$$G_X'(t)=\sum_{k\geq 1} x_k e^{t x_k} p_k$$

$$G_X'(0)=\sum x_k p_k=M(X)$$

$$G_X''(t)=\sum x_k^2 e^{t x_k} p_k$$

$$G_X''(0)=\sum x_k^2 p_k=M(X^2)$$

$$G_X^{(k)}(0)=M(X^k)$$

valabil pt $\forall t$ cand continuu

$$\varphi_X(t)=\sum e^{itx_k} p_k$$

$$\varphi_X'(t)=\sum_{k\geq 1} ix_k e^{itx_k} p_k \Rightarrow$$

$$\varphi_X''(t)=i^2\sum x_k^2 e^{itx_k} p_k$$

$$\varphi_X^{(k)}(0)=i^k\cdot M(X^k)$$

$$\varphi_X'(0)=iM(X)$$

$$\varphi_X''(0)=i^2M(X^2)=-M(X^2)$$

$$M(X^k)=\frac{1}{i^k}\cdot\varphi_X^{(k)}(0)$$

$$X:\begin{pmatrix} 0 & 1 & \dots & k & \dots \\ \frac{1}{1+a} & \frac{a}{(1+a)^2} & \dots & \frac{a^k}{(1+a)^{k+1}} & \end{pmatrix}, \quad a>0$$

Să det. fc. generatoare, fc. caract, media, dispersia

Sol $G_X(t) = \sum_{k\geq 0} e^{tk} \cdot P(X=k) = \sum_{k\geq 0} e^{tk} \frac{a^k}{(1+a)^{k+1}} =$

$= \frac{1}{1+a} \sum_{k\geq 0} \left(e^t \cdot \frac{a}{1+a}\right)^k = \frac{1}{1+a} \left[\sum_{k\geq 0} x^k\right] = \frac{1}{1+a} \cdot \frac{1}{1-x} =$

$1+q+q^2+\dots \quad 0<|q| = \left|e^t \frac{a}{1+a}\right| < 1$ (!) dacă

$= \frac{1}{1-q}$ $\quad e^t < \frac{a+1}{a}$

$t < \ln \frac{a+1}{a}$

$= \frac{1}{1+a} \cdot \frac{1}{1-e^t \frac{a}{1+a}} = \frac{1}{1+a} \cdot \frac{1+a}{1+a-ae^t} = \frac{1}{1+a-e^t a}$

fc. caract $\varphi_X(t) = \sum e^{itk} P(X=k) = \dots$

$= \frac{1}{1+a-e^{it}a}$

aici $|e^{it}| = |\cos t + i \sin t| = \sqrt{\cos^2 t + \sin^2 t} = 1.$

$\left|e^{it} \frac{a}{1+a}\right| = \underbrace{\left|e^{it}\right|}_{1} \left|\frac{a}{a+1}\right| < 1.$

avantajul fc. caracteristice

Să calc. val medie / dispers

$M(X^k) = \frac{1}{i^k} \varphi_X^{(k)}(0)$

formal $\varphi_X(t)$

$M(X) = \frac{1}{i} \varphi_X'(0)$

$\varphi' = \frac{0 - 1(-iae^{it})}{(1+a-e^{it}a)^2}$

$\varphi_X'(0) = \frac{ia e^0}{(1+a-e^0 a)^2} = ia$

$M(X) = \frac{1}{i} \cdot ia = a$

$D^2(X) = M(X^2) - (M(X))^2$

$M(X^2) = \frac{1}{i^2} \varphi_X''(0) = -\varphi_X''(0)$

$\varphi_X'(t) = \frac{ia e^{it}}{(1+a-ae^{it})^2}$ // $\varphi_X''(t) = ia \frac{ie^{it}(\ ) - e^{it} 2(\ )(-aie^{it})}{(1+a-ae^{it})^4}$

$= ia \frac{ie^{it} + 2aie^{it} \cdot e^{it}}{(1+a-ae^{it})^3}$; $\varphi_X''(0) = ia \frac{i+2ai}{1}$

$= i^2 a(1+2a) = -a(1+2a) \Rightarrow$

$M(X^2) = -\varphi_X''(0) = a(1+2a) \Rightarrow$

$D^2(X) = a(1+2a) - a^2 = a + 2a^2 - a^2 =$

$= a^2 + a = a(a+1)$

Ex $X_k : \begin{pmatrix} -\frac{1}{2^k} & \frac{1}{2^k} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$ $k = 1, 2, \dots, n$, independente

Se cere fc. caract pt $X = \sum_{k=1}^{n} X_k$.

Sol

$\varphi_X(t) = M(e^{itX}) = M\left(e^{it(\sum_{k=1}^{n} X_k)}\right) = M\left(e^{\sum_{k=1}^{n} (itX_k)}\right)$

$= M(e^{itX_1} \cdot e^{itX_2} \cdot \dots \cdot e^{itX_n})$ $X_1, \dots, X_n$ indep

$= M(e^{itX_1}) \cdot M(e^{itX_2}) \cdot \dots \cdot M(e^{itX_n}) = \varphi_{X_1}(t) \varphi_{X_2}(t) \dots \varphi_{X_n}(t)$

$= \prod_{k=1}^{n} \varphi_{X_k}(t)$

$$\varphi_{X_k}(t) = M\left(e^{itX_k}\right) = e^{-\frac{1}{2^k}\cdot it}\cdot\frac{1}{2} + e^{\frac{1}{2^k}\cdot it}\cdot\frac{1}{2} =$$

$$= \frac{1}{2}\left(e^{-\frac{it}{2^k}} + e^{\frac{it}{2^k}}\right) = \cos\frac{t}{2^k}.$$

$$\begin{cases} \cos\alpha + i\sin\alpha = e^{i\alpha} \\ \cos\alpha - i\sin\alpha = e^{-i\alpha} \end{cases}$$

$$\cos\alpha = \frac{1}{2}\left(e^{i\alpha} + e^{-i\alpha}\right)$$

$$\sin\alpha = \frac{1}{2i}\left(e^{i\alpha} - e^{-i\alpha}\right)$$

$$\Longrightarrow \varphi_X(t) = \prod_{k=1}^{n} \cos\frac{t}{2^k} =$$

$$= \cos\frac{t}{2}\cdot\cos\frac{t}{2^2}\cdot\ldots\cdot\cos\frac{t}{2^n} =$$

$$= \frac{\sin t}{2^n\cdot\sin\frac{t}{2^n}}$$

$$\left(\lim_{n\to\infty}\varphi_X(t) = \lim_{n\to\infty}\frac{\sin t}{2^n\cdot\sin\frac{t}{2^n}} =\right.$$

$$= \frac{\sin t}{t}\cdot\lim_{n\to\infty}\frac{\frac{t}{2^n}}{\sin\frac{t}{2^n}} = \frac{\sin t}{t}.$$

$$\lim_{n\to\infty}\frac{t}{2^n} = 0$$

$$\lim_{x\to 0}\frac{\sin x}{x} = 1$$

$$\lim_{x\to 0}\frac{x}{\sin x} = 1$$

formula generala

$$\sin 2t = 2\sin t\cos t$$

inmultim

$$\sin t = 2\sin\frac{t}{2}\cos\frac{t}{2}$$

$$\sin\frac{t}{2} = 2\sin\frac{t}{2^2}\cos\frac{t}{2^2}$$

$$\sin\frac{t}{2^2} = 2\sin\frac{t}{2^3}\cos\frac{t}{2^3}$$

$$\vdots$$

$$\sin\frac{t}{2^{n-1}} = 2\sin\frac{t}{2^n}\cos\frac{t}{2^n}$$

$$\sin t = 2^n\cdot\cos\frac{t}{2}\cos\frac{t}{2^2}\ldots\cos\frac{t}{2^n}\cdot\sin\frac{t}{2^n}$$

Ex Se dă $\varphi_X(t) = \frac{1}{8}(1+e^{it})^3$.

Să se det $M(X)$, $D^2(X)$ și să se deducem repartiția v.a. X; Să scriem fc. de repart

$\varphi_X(t) = \frac{1}{8}(1 + 3e^{it} + 3e^{2it} + e^{3it})$

$= \frac{1}{8} \cdot e^{i \cdot 0 \cdot t} + \frac{3}{8} e^{i \cdot 1 \cdot t} + \frac{3}{8} e^{2t \cdot i} + \frac{1}{8} e^{3ti}$

$X : \begin{pmatrix} & & & \\ \frac{1}{8} & \frac{3}{8} & \frac{3}{8} & \frac{1}{8} \end{pmatrix}$